Spectral Simulation Report

Automated Analysis

July 4, 2025

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1 Sample: 14615

1.1 Simulation Parameters

Parameter	Value
Temperature (T)	7929 K
Electron Density (n_e)	1.87×10^{17} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	40.47

Table 1: Simulation parameters and additional metrics for sample 14615.

1.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	4.39	279.0
Ar	0.54	441.0
C	6.56	221.0
Ca	8.41	150.0
Cl	3.48	12.0
Co	13.35	1402.0
Cr	8.01	397.0
Fe	6.60	2673.0
Mg	12.77	323.0
Mn	13.25	706.0
N	0.43	396.0
Na	2.71	330.0
Ni	3.98	478.0
O	1.42	83.0
S	10.48	25.0
Si	2.95	205.0
Ti	0.67	1181.0
background	N/A	497.0
Total Positions		9799.0

Table 2: Composition analysis for sample 14615.

1.3 Spectral Plot

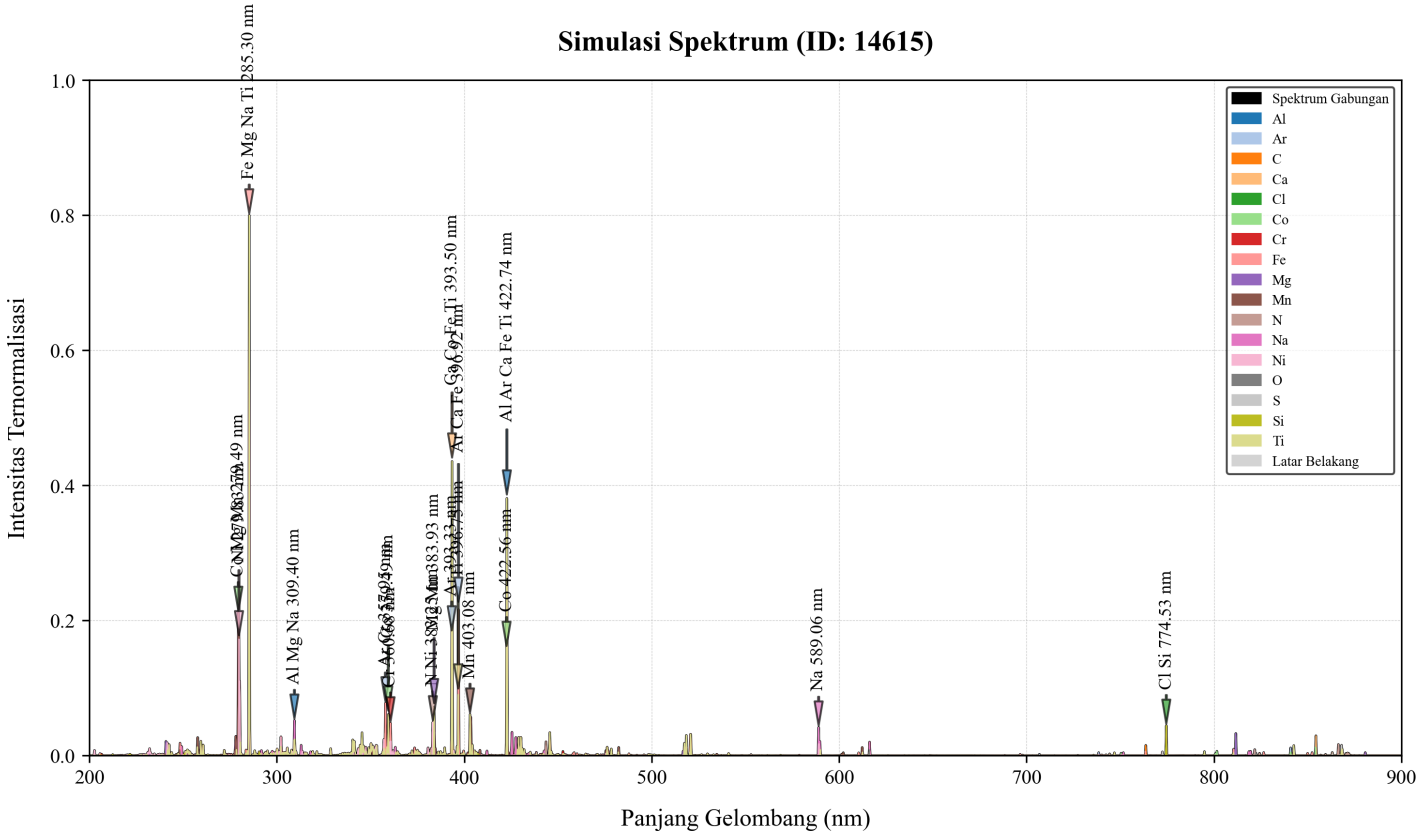

Figure 1: Spectral simulation plot for sample 14615 with multi-label encoding and detailed composition analysis.

2 Sample: 5302

2.1 Simulation Parameters

Parameter	Value
Temperature (T)	9848 K
Electron Density (n_e)	1.75×10^{14} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	39.74

Table 3: Simulation parameters and additional metrics for sample 5302.

2.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	6.32	451.0
Ar	0.00	0.0
C	0.00	0.0
Ca	7.75	150.0
Cl	5.07	35.0
Co	0.00	0.0
Cr	9.01	397.0
Fe	0.00	0.0
Mg	17.43	349.0
Mn	10.07	706.0
N	15.35	428.0
Na	1.05	315.0
Ni	7.87	474.0
O	4.60	160.0
S	11.68	228.0
Si	0.82	217.0
Ti	2.97	1184.0
background	N/A	1299.0
Total Positions		6393.0

Table 4: Composition analysis for sample 5302.

2.3 Spectral Plot

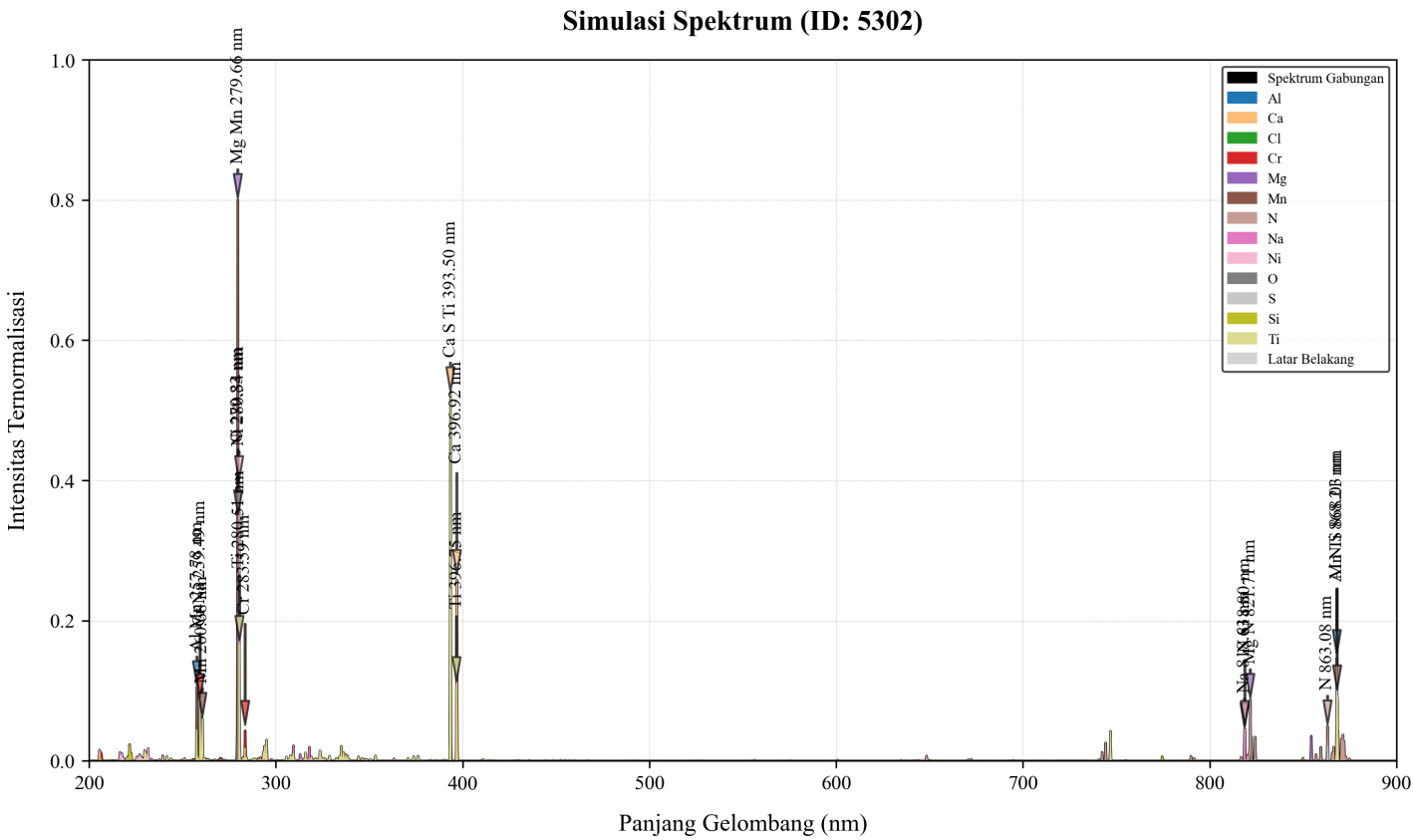

Figure 2: Spectral simulation plot for sample 5302 with multi-label encoding and detailed composition analysis.

3 Sample: 9016

3.1 Simulation Parameters

Parameter	Value
Temperature (T)	9545 K
Electron Density (n_e)	3.59×10^{17} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	34.80

Table 5: Simulation parameters and additional metrics for sample 9016.

3.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	1.00	419.0
Ar	1.28	721.0
C	0.00	0.0
Ca	3.97	150.0
Cl	6.37	18.0
Co	0.00	0.0
Cr	16.22	397.0
Fe	0.00	0.0
Mg	0.00	0.0
Mn	6.14	706.0
N	0.00	0.0
Na	6.04	330.0
Ni	12.11	478.0
O	11.93	189.0
S	8.07	190.0
Si	16.26	223.0
Ti	10.61	1186.0
background	N/A	1360.0
Total Positions		6367.0

Table 6: Composition analysis for sample 9016.

3.3 Spectral Plot

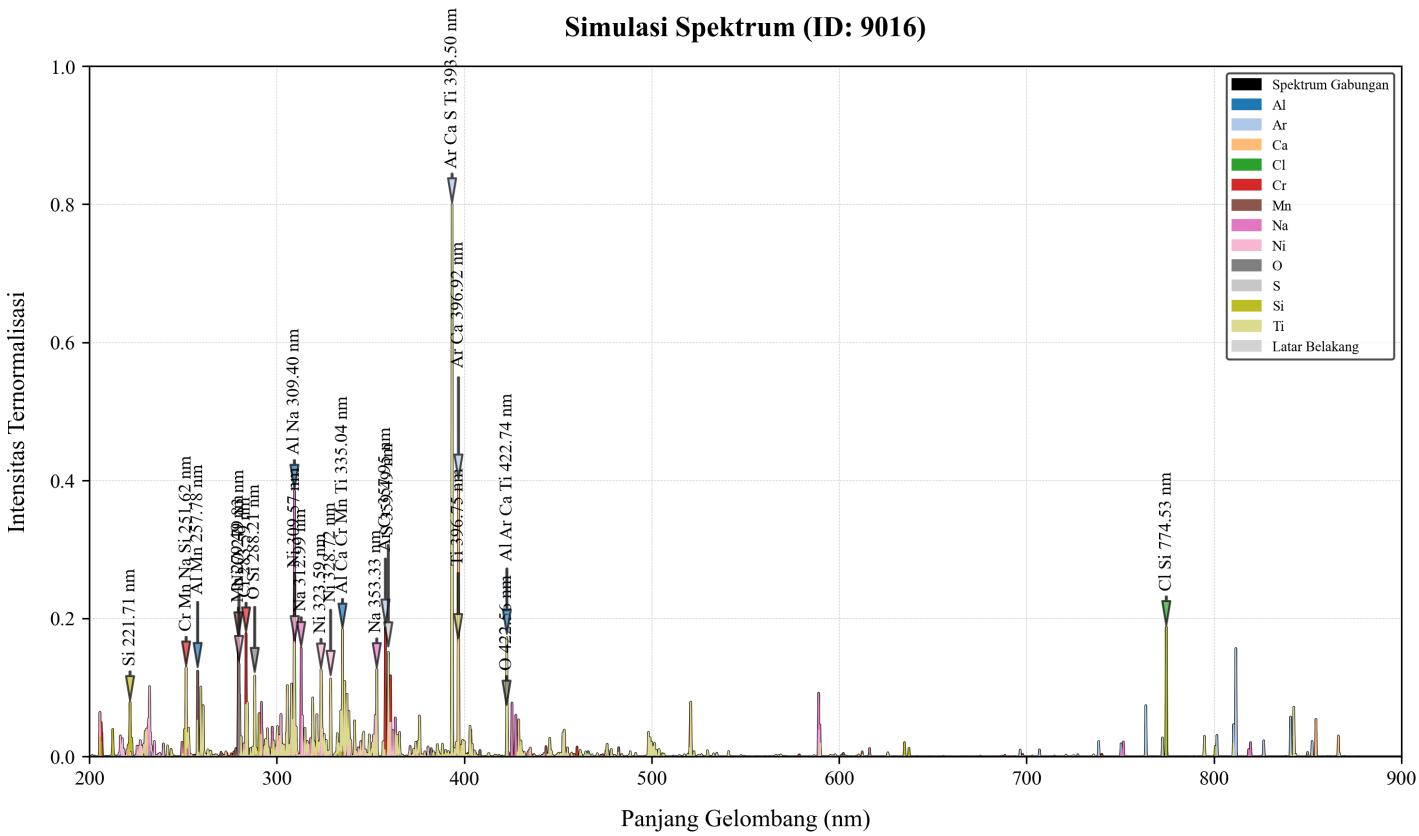

Figure 3: Spectral simulation plot for sample 9016 with multi-label encoding and detailed composition analysis.

4 Sample: 5605

4.1 Simulation Parameters

Parameter	Value
Temperature (T)	12 374 K
Electron Density (n_e)	2.66×10^{16} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	41.25

Table 7: Simulation parameters and additional metrics for sample 5605.

4.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	0.00	0.0
Ar	6.69	727.0
C	8.31	623.0
Ca	6.14	150.0
Cl	6.04	35.0
Co	4.15	1404.0
Cr	11.79	397.0
Fe	0.00	0.0
Mg	11.63	356.0
Mn	10.29	706.0
N	0.00	0.0
Na	7.78	325.0
Ni	9.85	478.0
O	0.00	0.0
S	7.36	228.0
Si	9.98	223.0
Ti	0.00	0.0
background	N/A	1095.0
Total Positions		6747.0

Table 8: Composition analysis for sample 5605.

4.3 Spectral Plot

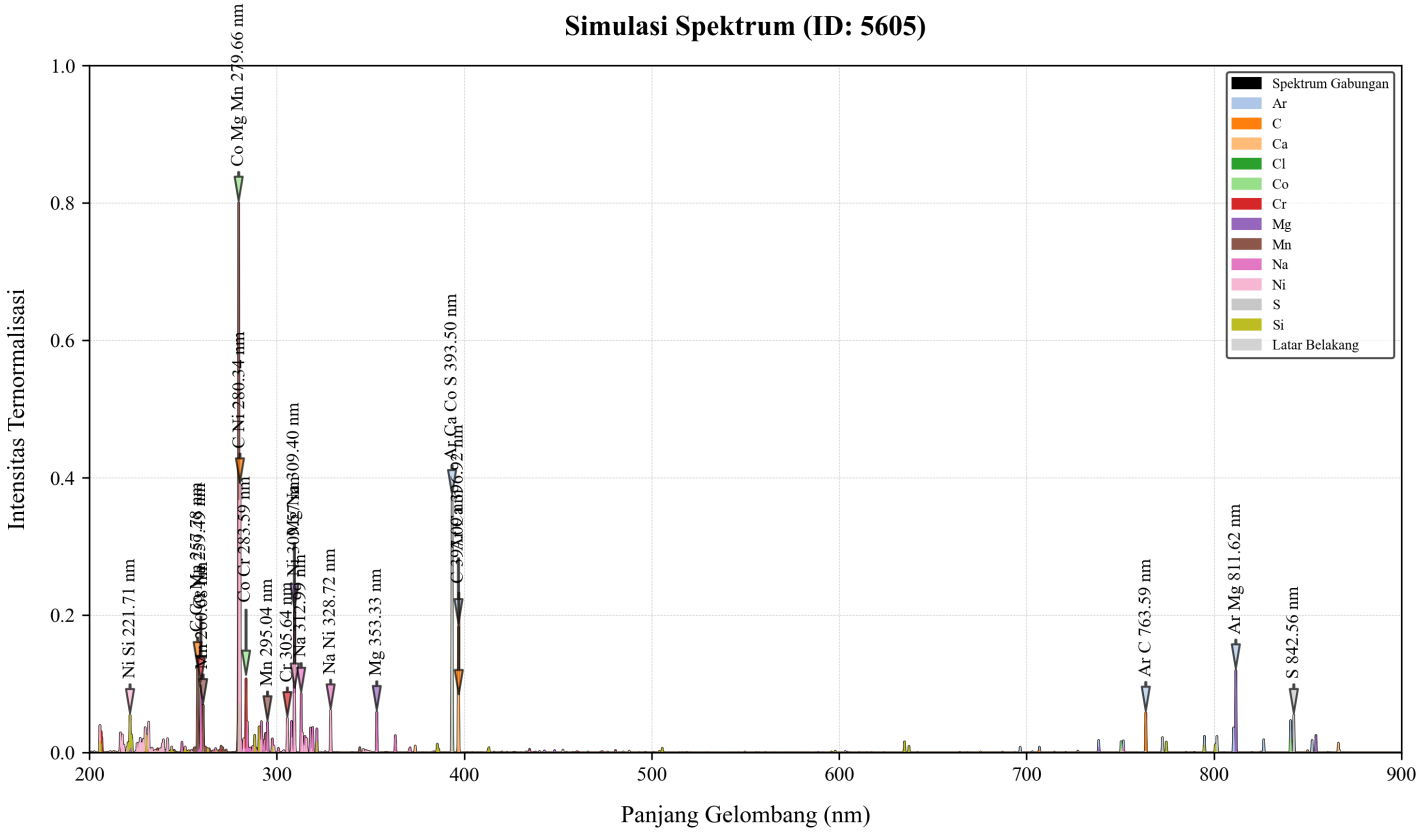

Figure 4: Spectral simulation plot for sample 5605 with multi-label encoding and detailed composition analysis.

5 Sample: 8712

5.1 Simulation Parameters

Parameter	Value
Temperature (T)	11 970 K
Electron Density (n_e)	4.75×10^{17} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	34.47

Table 9: Simulation parameters and additional metrics for sample 8712.

5.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	9.22	462.0
Ar	9.72	727.0
C	2.25	327.0
Ca	7.75	150.0
Cl	11.89	33.0
Co	5.65	1404.0
Cr	4.19	397.0
Fe	9.43	2761.0
Mg	0.00	0.0
Mn	6.11	706.0
N	0.00	0.0
Na	12.49	330.0
Ni	10.56	478.0
O	0.00	0.0
S	3.88	228.0
Si	6.85	223.0
Ti	0.00	0.0
background	N/A	532.0
Total Positions		8758.0

Table 10: Composition analysis for sample 8712.

5.3 Spectral Plot

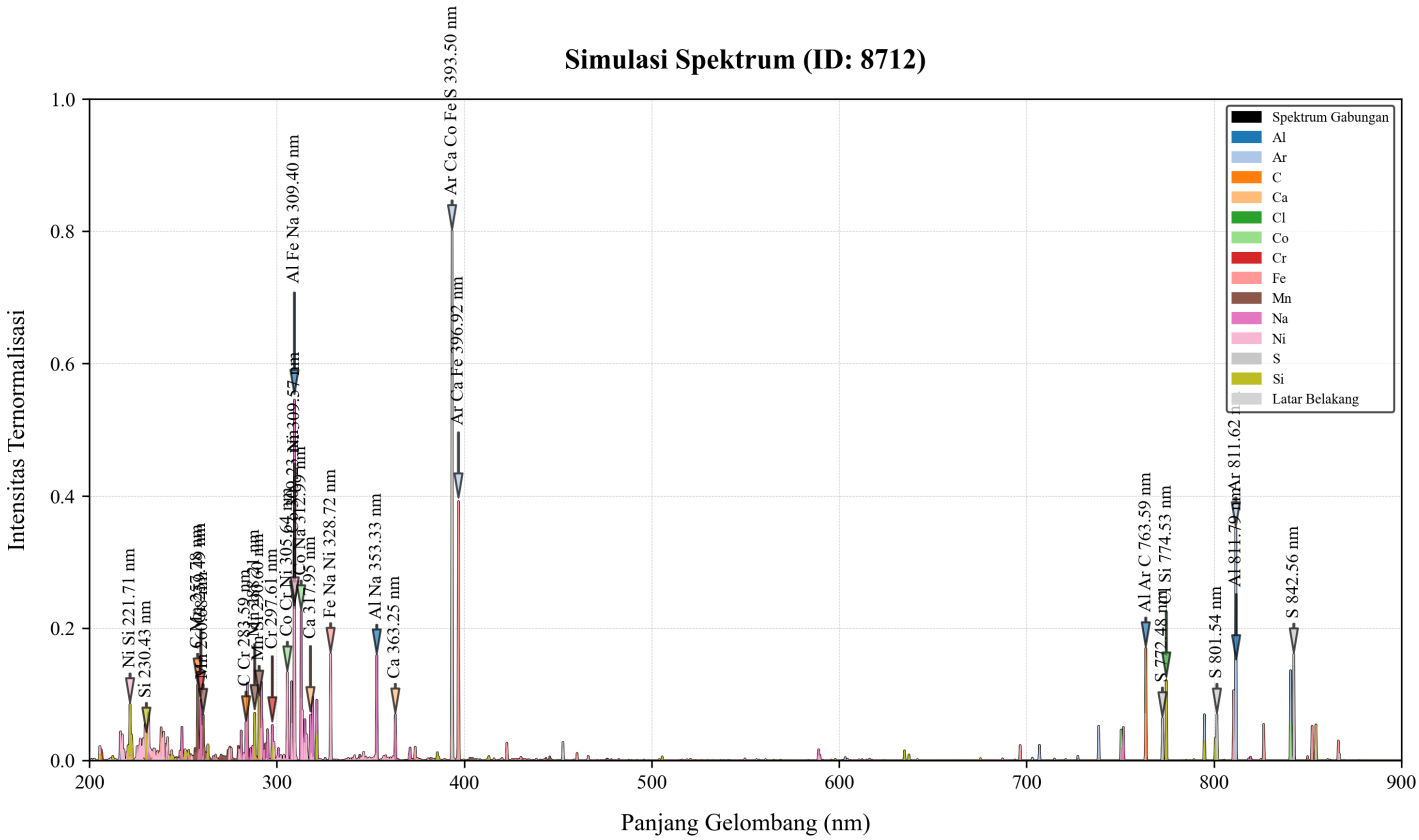

Figure 5: Spectral simulation plot for sample 8712 with multi-label encoding and detailed composition analysis.

6 Sample: 5466

6.1 Simulation Parameters

Parameter	Value
Temperature (T)	10 152 K
Electron Density (n_e)	2.54×10^{14} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	41.80

Table 11: Simulation parameters and additional metrics for sample 5466.

6.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	6.06	456.0
Ar	0.51	727.0
C	12.12	300.0
Ca	8.17	150.0
Cl	1.45	35.0
Co	7.53	1404.0
Cr	9.36	397.0
Fe	0.00	0.0
Mg	0.71	337.0
Mn	11.04	706.0
N	2.06	405.0
Na	10.53	322.0
Ni	1.01	469.0
O	4.30	165.0
S	4.59	226.0
Si	8.92	223.0
Ti	11.64	1189.0
background	N/A	726.0
Total Positions		8237.0

Table 12: Composition analysis for sample 5466.

6.3 Spectral Plot

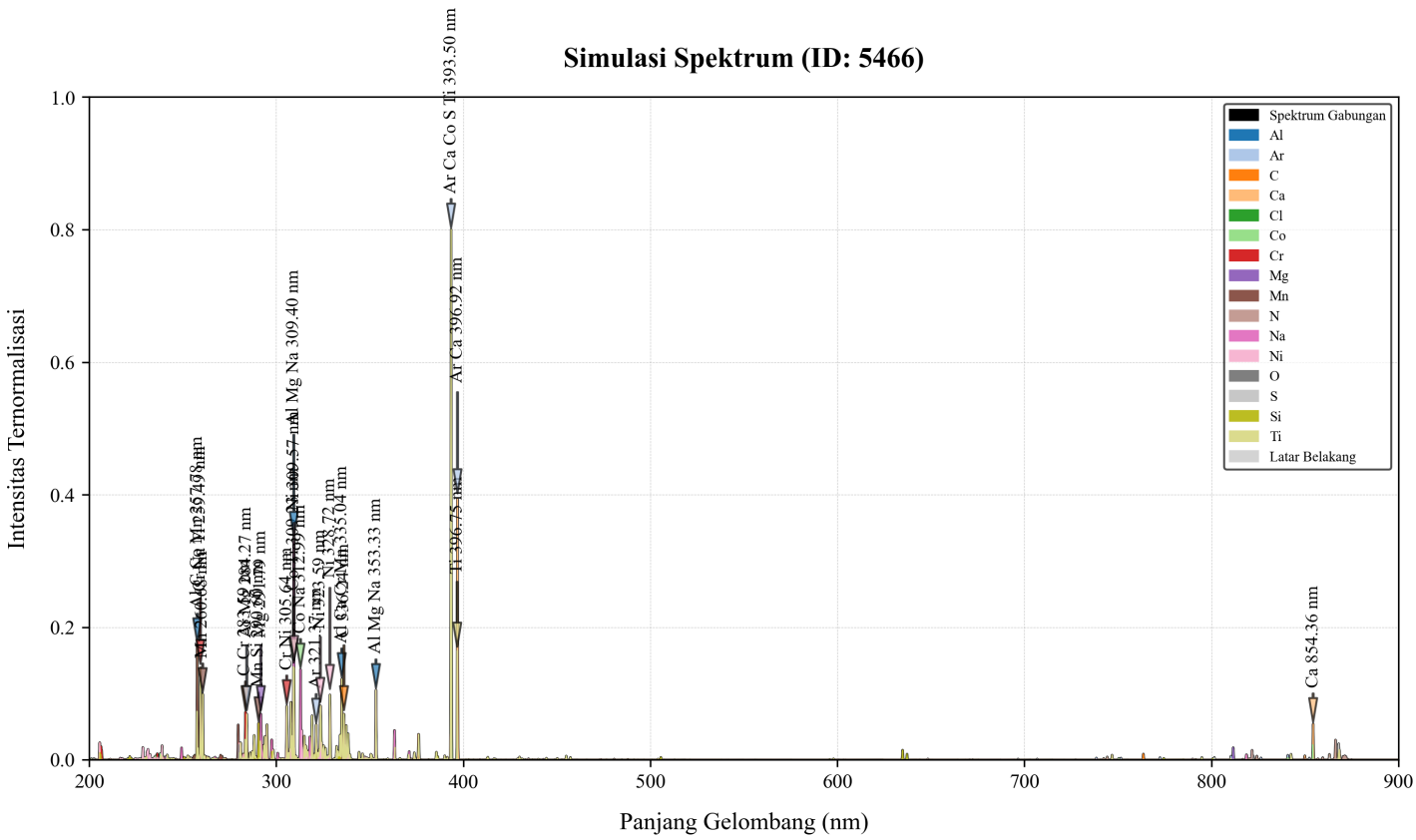

Figure 6: Spectral simulation plot for sample 5466 with multi-label encoding and detailed composition analysis.

7 Sample: 6992

7.1 Simulation Parameters

Parameter	Value
Temperature (T)	6818 K
Electron Density (n_e)	3.43×10^{15} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	35.88

Table 13: Simulation parameters and additional metrics for sample 6992.

7.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	0.00	0.0
Ar	0.00	0.0
C	0.00	0.0
Ca	11.81	150.0
Cl	1.49	12.0
Co	2.01	1026.0
Cr	3.21	397.0
Fe	1.19	2266.0
Mg	14.55	256.0
Mn	8.82	697.0
N	8.75	396.0
Na	12.97	326.0
Ni	1.80	478.0
O	6.73	59.0
S	15.92	25.0
Si	10.36	218.0
Ti	0.40	1181.0
background	N/A	962.0
Total Positions		8449.0

Table 14: Composition analysis for sample 6992.

7.3 Spectral Plot

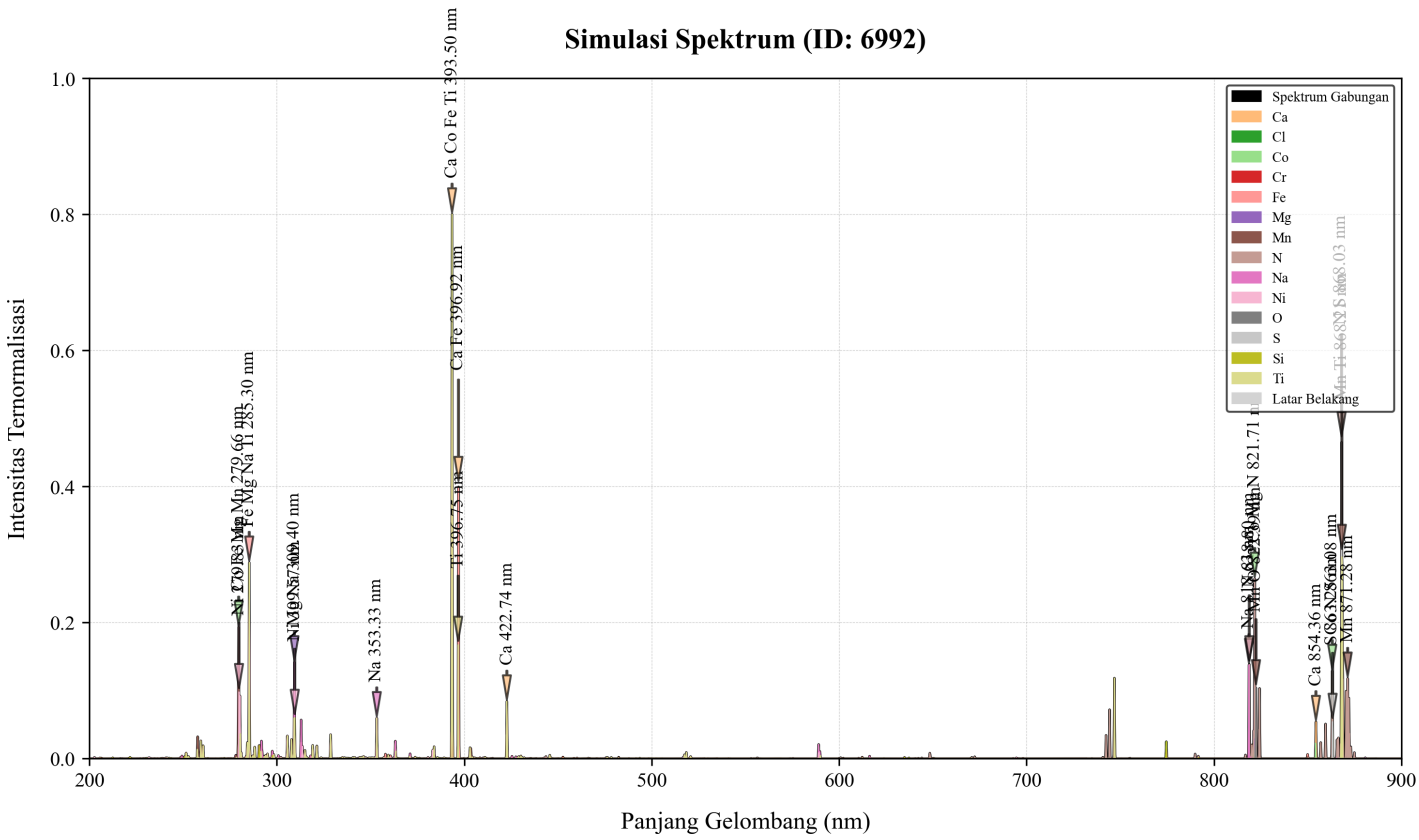

Figure 7: Spectral simulation plot for sample 6992 with multi-label encoding and detailed composition analysis.

8 Sample: 8367

8.1 Simulation Parameters

Parameter	Value
Temperature (T)	10 960 K
Electron Density (n_e)	2.98×10^{17} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	41.66

Table 15: Simulation parameters and additional metrics for sample 8367.

8.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	5.71	462.0
Ar	0.78	727.0
C	9.23	284.0
Ca	0.39	150.0
Cl	11.50	29.0
Co	0.00	0.0
Cr	7.14	397.0
Fe	3.69	2733.0
Mg	10.84	359.0
Mn	10.62	706.0
N	3.77	399.0
Na	7.21	330.0
Ni	9.91	478.0
O	9.82	196.0
S	3.46	228.0
Si	1.41	218.0
Ti	4.53	1186.0
background	N/A	494.0
Total Positions		9376.0

Table 16: Composition analysis for sample 8367.

8.3 Spectral Plot

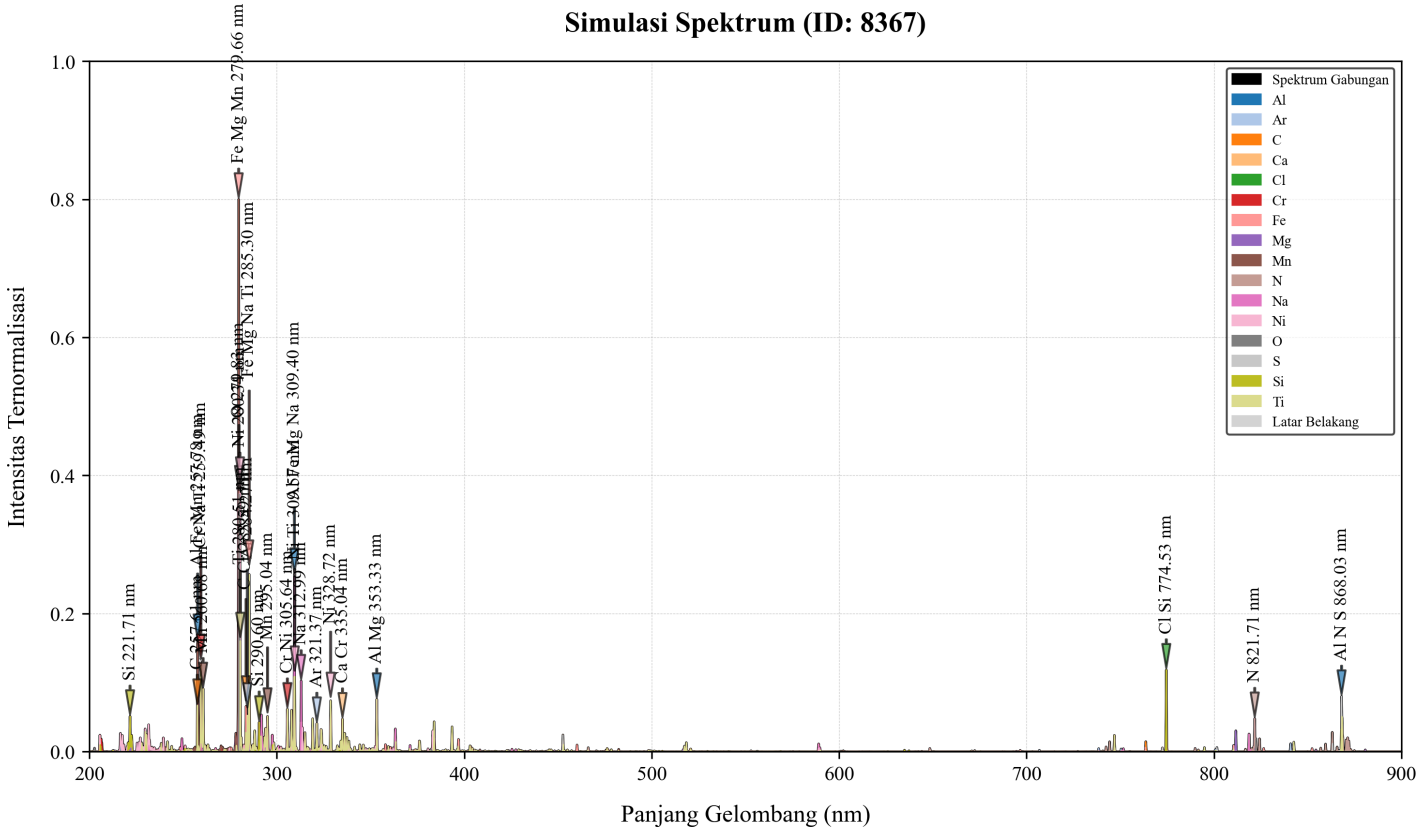

Figure 8: Spectral simulation plot for sample 8367 with multi-label encoding and detailed composition analysis.

9 Sample: 16029

9.1 Simulation Parameters

Parameter	Value
Temperature (T)	9343 K
Electron Density (n_e)	4.23×10^{16} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	27.72

Table 17: Simulation parameters and additional metrics for sample 16029.

9.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	14.11	451.0
Ar	7.56	727.0
C	13.97	247.0
Ca	5.15	150.0
Cl	3.35	22.0
Co	1.31	1404.0
Cr	3.07	397.0
Fe	0.00	0.0
Mg	3.86	347.0
Mn	1.16	706.0
N	4.39	396.0
Na	14.05	330.0
Ni	8.84	478.0
O	1.31	139.0
S	4.19	199.0
Si	13.67	223.0
Ti	0.00	0.0
background	N/A	889.0
Total Positions		7105.0

Table 18: Composition analysis for sample 16029.

9.3 Spectral Plot

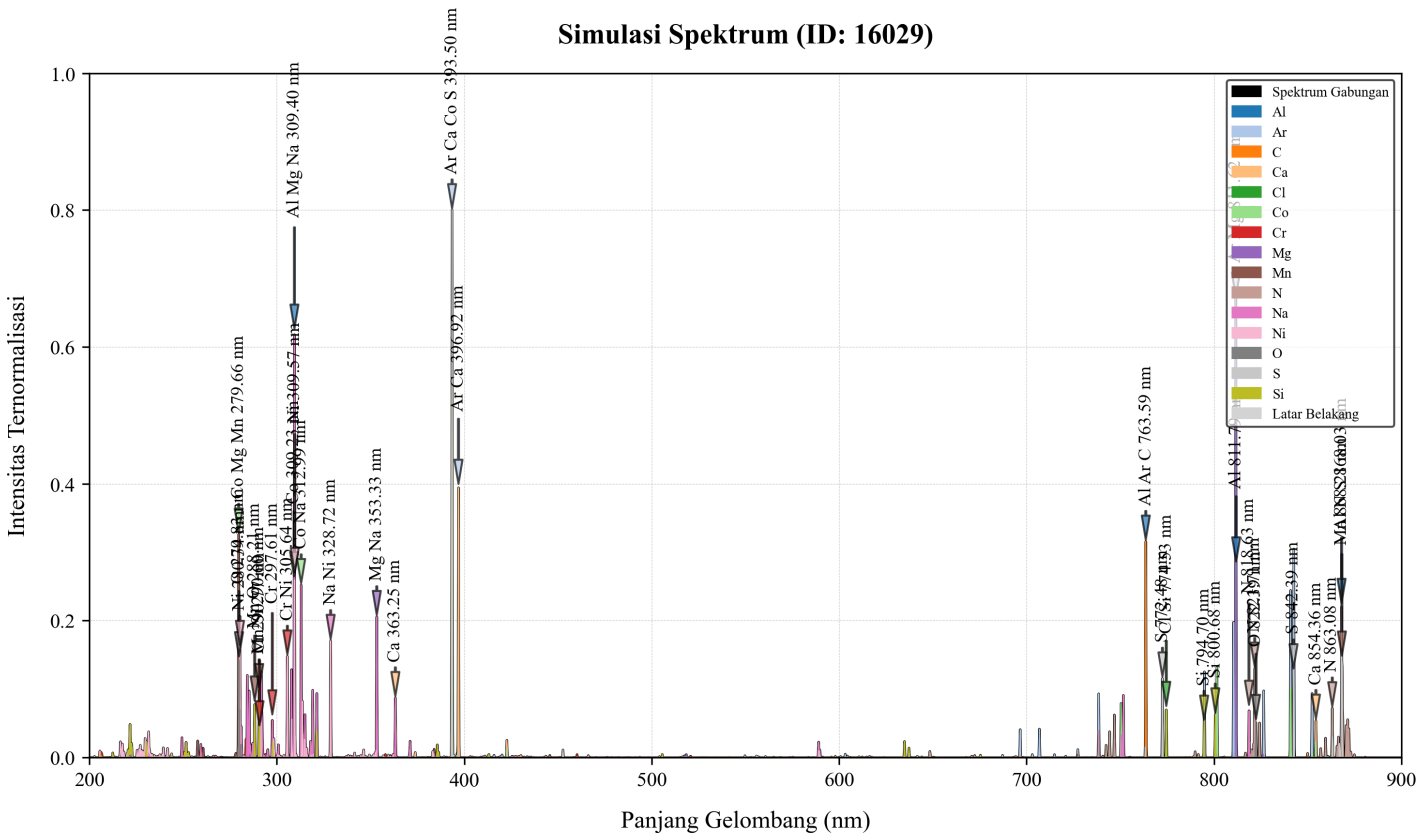

Figure 9: Spectral simulation plot for sample 16029 with multi-label encoding and detailed composition analysis.

10 Sample: 1076

10.1 Simulation Parameters

Parameter	Value
Temperature (T)	6313 K
Electron Density (n_e)	2.15×10^{15} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	39.10

Table 19: Simulation parameters and additional metrics for sample 1076.

10.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	8.84	161.0
Ar	10.90	418.0
C	13.67	207.0
Ca	4.55	150.0
Cl	2.59	12.0
Co	0.00	0.0
Cr	0.51	397.0
Fe	0.00	0.0
Mg	2.76	173.0
Mn	11.03	684.0
N	4.35	396.0
Na	5.04	325.0
Ni	0.60	478.0
O	8.87	28.0
S	11.15	25.0
Si	8.09	207.0
Ti	7.03	1181.0
background	N/A	1323.0
Total Positions		6165.0

Table 20: Composition analysis for sample 1076.

10.3 Spectral Plot

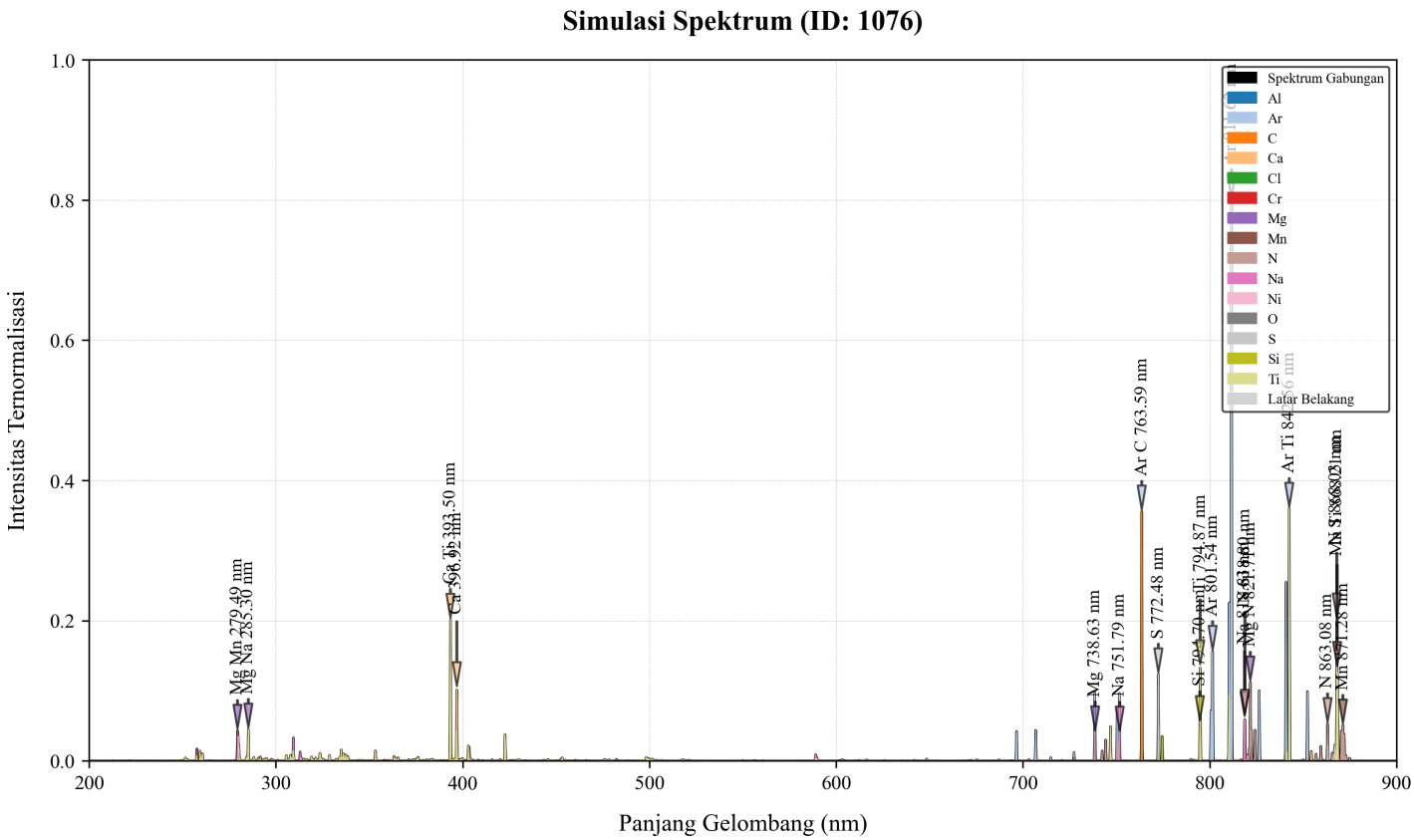

Figure 10: Spectral simulation plot for sample 1076 with multi-label encoding and detailed composition analysis.